

Autonomous Multi-Sensor Fire Detection Node Using Sensor Fusion and TinyML

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Abstract

This thesis presents the development of an intelligent edge-deployed fire detection system...

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1 Introduction

1.1 Background

1.1.1 Significance of Fire Detection and False Alarm Issues

Fire-related incidents continue to pose a severe threat to global life safety, infrastructure, and ecological stability. In 2024, the United States reported approximately 1.39 million fires, which resulted in 3,920 civilian deaths and an estimated \$19.1 billion in direct property damage. Beyond the residential domain, wildland fires have intensified due to shifting climate patterns; the 2024–2025 fire season recorded carbon emissions of approximately 2.2 Pg C, which is 9% above the historical average. These extreme wildfire events, particularly in the North American boreal forests and South American wetlands, underscore the urgent need for autonomous, rapid-response detection systems capable of operating in diverse environmental conditions (Khan & others, 2025).

However, the efficacy of modern fire safety infrastructure is severely undermined by the prevalence of false alarms, often referred to as nuisance alarms. In the United Kingdom, Home Office data for the 2020/21 period revealed that 98% of all incidents triggered by automatic fire detection systems were false alarms, with only 2% resulting from actual fires. Approximately 90% of these false positives are attributed to faulty apparatus or environmental interference rather than malicious intent. This “cry wolf” effect leads to dangerous occupant desensitization, where individuals may ignore or disable life-saving equipment due to frequent erroneous triggers (Pathan & others, 2024).

The economic ramifications of these inaccuracies are profound. In the UK, false fire alarms cause annual losses exceeding £1 billion due to lost productivity and business disruption, with the average cost per instance estimated at £2,900. Similarly, regional reports from New South Wales, Australia, indicate that false alarms account for nearly 37% of total fire department responses, straining public resources and delaying deployment to genuine emergencies (BRE Group, 2019). Consequently, reducing the false alarm rate is a critical requirement for maintaining the integrity of fire services and public trust in safety technology (Western Sydney University, 2019).

1.1.2 Limitations of Traditional Fire Detectors vs. Intelligent Edge Nodes

Traditional fire detection systems primarily rely on single-parameter sensing, such as ionization or photoelectric smoke detection. While effective in specific scenarios, these sensors lack the granularity to distinguish between genuine combustion and common household or industrial non-fire aerosols. Photoelectric sensors are notoriously prone to interference from water vapor (steam), dust, and cooking fumes, while ionization sensors frequently trigger in response to high humidity or minor culinary activities. These single-modality systems operate on static thresholding, which fails to account for the complex, time-varying signatures of early-stage fires or the cross-correlation between temperature, gas concentrations, and light radiation (Singh, 2024).

The transition toward intelligent edge nodes offers a paradigm shift in detection reliability. By integrating a multi-sensor suite—incorporating Volatile Organic Compounds (VOC), Carbon Monoxide (CO), Infrared (IR) flame detection, and environmental variables (temperature and humidity)—a node can capture the multi-dimensional fingerprint of a fire event. Modern microcontrollers, such as the Cortex-M4 class powering the Arduino UNO R4 WiFi, enable the deployment of Tiny Machine Learning (TinyML) models directly at the

sensing site. This allows for real-time sensor fusion where the system processes high-frequency data (e.g., 10 Hz) across temporal windows (typically 10 s) to identify patterns that static thresholds overlook (Perez & Smith, 2023).

Unlike traditional sensors, these intelligent nodes utilize local inference to classify environments into three distinct states: fire, no_fire, and false_alarm. This classification framing significantly reduces latency by eliminating the need for cloud-based processing while preserving battery life through efficient on-device computation. Research indicates that TinyML-driven multi-sensor fusion can achieve detection accuracies exceeding 98% while concurrently reducing false alarm rates to below 1%, effectively overcoming the physical limitations of legacy hardware (NJCCS, 2024).

1.2 Problem Statement

1.2.1 Background and Problem Statement

The prevalence of false alarms in traditional fire detection systems remains a critical challenge for emergency response efficiency and public safety. Statistics from the United Kingdom Home Office indicate that approximately 98% of automatic fire alarm (AFA) incidents confirmed between 2020 and 2021 were false alarms, with 90% of these attributed to faulty or improperly maintained apparatus (International Fire and Safety Journal, 2022). Similar trends are observed in high-density urban infrastructure; for instance, the Taipei Metro reported a 94% false alarm rate between 2019 and 2024, where only 28 of 484 activations involved actual fire events (Anonymous, 2025). These frequent unwanted alarms lead to significant economic losses due to interrupted operations, the exhaustion of fire service resources, and a phenomenon known as “alarm fatigue,” where occupants begin to ignore signals, thereby increasing the risk of casualties during real fire events (Wang et al., 2023).

The high susceptibility to false positives stems largely from the limitations of traditional single-sensor detectors, which typically rely on a single physical property such as ionization or light scattering. These devices struggle to differentiate between combustion-derived smoke and non-fire aerosols like steam, dust, or cooking fumes (Hatip & Zayood, 2024). Single-sensor architectures are fundamentally limited in their ability to classify complex environmental patterns, as smoke particles vary significantly in composition and density (Hatip & Zayood, 2024). Furthermore, traditional rule-based algorithms often lack the flexibility to adapt to varying ambient conditions, resulting in either high false alarm rates or delayed response times (Edge Impulse, 2026).

To address these vulnerabilities, there is a growing need for intelligent multi-sensor fusion systems deployed at the network edge. Recent advancements in Tiny Machine Learning (TinyML) enable the execution of complex neural networks on resource-constrained microcontrollers, such as the ARM Cortex-M4 (Moosmann et al., 2024). By integrating heterogeneous data from gas, thermal, and optical sensors, these systems can identify cross-sensor correlations that are characteristic of true fire signatures (Wang et al., 2023). However, a significant gap remains in developing low-power, cost-effective nodes that can provide three-class classification—distinguishing between “Fire,” “No Fire,” and specific “False Alarm” triggers—to minimize unnecessary emergency dispatches (Edge Impulse, 2026).

1.2.2 Scope and Limitations

This research focuses on the development and validation of a stationary “Autonomous Node” for indoor fire detection, utilizing an Arduino-based hardware platform. The system is

designed to provide continuous 24/7 monitoring of specific indoor zones by fusing data from a multi-modal sensor array, including Micro-Electro-Mechanical Systems (MEMS) smoke, volatile organic compound (VOC), and carbon monoxide (CO) sensors, alongside infrared (IR) flame detection and environmental temperature/humidity sensing (Rasimmax, 2024). Unlike mobile robotic platforms or Unmanned Aerial Systems (UAS) used for forest fire patrol, the stationary node is optimized for fixed installation in residential or commercial environments where high-fidelity, localized monitoring is required (Meleti & Razi, 2024).

The scope of this project is strictly limited to indoor fire detection scenarios. The sensor suite is selected to detect common indoor fire signatures; however, the detection range is limited by the physical sensitivity of the MEMS gas sensors and the line-of-sight requirements of the IR flame sensor (Edge Impulse, 2026). While the system leverages an Arduino UNO R4 WiFi for processing and communication, the research emphasizes TinyML model efficiency, specifically targeting a 32-bit Cortex-M4 architecture with limited SRAM (Moosmann et al., 2024). The system does not address active fire suppression or complex multi-node mesh networking, focusing instead on the accuracy of the individual node’s classification capabilities.

Furthermore, while the node can operate on battery power, long-term deployment strategies for energy harvesting or extreme low-power sleep modes are considered secondary to the primary objective of false alarm reduction through sensor fusion.

1.3 Research Questions

1.3.1 Sensor Fusion for False Alarm Discrimination

The primary challenge in modern fire safety is the prevalence of unwanted alarms triggered by non-combustion aerosols, such as steam, cooking fumes, or cleaning sprays, which frequently deactivate automated detection systems (Müller et al., 2024). While traditional single-modality detectors struggle with these “nuisance” triggers, multi-sensor data fusion—integrating particulate smoke, Volatile Organic Compounds (VOC), Carbon Monoxide (CO), and infrared flame signatures—offers a mathematically robust framework for optimal sensor integration under uncertainty (Zheng et al., 2022). Empirical evidence suggests that multi-sensor frameworks can reduce false alarms by up to 73% compared to conventional single-sensor systems (Zheng et al., 2022). This study evaluates whether a specific array of MEMS-based gas sensors and infrared detectors can reliably distinguish between active combustion and ambient environmental variances. Consequently, the first research question asks: *Can a multi-sensor fusion approach effectively discriminate between real fire events, normal ambient conditions, and common indoor false alarm triggers?* This inquiry specifically focuses on the separability of multi-modal data in a stationary node context, targeting the identification of unique combustion signatures that remain distinct from household aerosols.

1.3.2 Real-Time TinyML Constraints on Cortex-M4 Platforms

Deploying sophisticated machine learning models on resource-constrained microcontrollers necessitates a rigorous balance between classification accuracy and execution speed (Das & others, 2025). The Cortex-M4 architecture, while widely adopted for its Digital Signal Processing (DSP) support, operates at lower clock frequencies than standard edge processors, such as the 48MHz Renesas RA4M1 utilized in the Arduino UNO R4 (Das & others, 2025). For fire detection systems to be considered safe for real-world application, inference latency must typically remain below 100ms to ensure near-instantaneous response times (Das & others, 2025). Existing research indicates that achieving such low latency often requires intensive model optimization, including quantization and pruning, to fit within the limited

RAM and Flash memory of embedded devices (Müller et al., 2024). This study seeks to determine the technical feasibility of running a fully integrated neural network on this low-power hardware without exceeding critical processing thresholds. Therefore, the second research question is: *Is TinyML deployment on a 48MHz Cortex-M4 microcontroller feasible for real-time fire detection while maintaining an inference latency of less than 100ms?*

1.3.3 Evaluation of Three-Class versus Binary Classification

Current fire detection literature is largely dominated by binary classification models that categorize events simply as “fire” or “no-fire” (Müller et al., 2024). Such binary frameworks often lack the granularity to handle complex non-fire scenarios, frequently grouping false alarm triggers with either the fire class (leading to false positives) or the no-fire class (leading to high variance in the “normal” baseline) (Müller et al., 2024). Advanced TinyML strategies now advocate for multi-class paradigms, where explicit “false alarm” labels allow the model to learn the specific statistical patterns of culinary activities or chemical sprays (Das & others, 2025). By providing the model with a dedicated class for nuisance triggers, it is hypothesized that the precision of the actual “fire” classification will improve significantly. This comparison is vital for understanding whether the increased complexity of a three-class model translates into tangible reliability gains in domestic and industrial environments. Thus, the third research question investigates: *How does three-class classification (fire/no_fire/false_alarm) compare to traditional binary classification in reducing false positive rates for intelligent fire detection nodes?*

1.4 Research Objectives and Contributions

1.4.1 Primary Research Objectives

The primary goal of this research is to develop and validate an autonomous sensing node capable of intelligent fire detection through multi-modal sensor fusion and edge-deployed machine learning. Traditional fire detection systems often suffer from high false alarm rates because they rely on single-modality thresholds, which fail to distinguish between combustion byproducts and common household aerosols (Müller et al., 2024). To address these limitations, this study pursues three core objectives. First, the research aims to design a robust sensor fusion framework that integrates particulate smoke, volatile organic compounds (VOC), carbon monoxide (CO), and infrared flame signatures. By leveraging the mathematical correlations between these distinct environmental variables, the system seeks to improve the reliability of fire detection in complex indoor environments where “nuisance” triggers are prevalent (Zheng et al., 2022).

The second objective focuses on the technical feasibility of deploying sophisticated neural networks on resource-constrained hardware. Specifically, this study evaluates the implementation of TinyML models on a 48MHz Cortex-M4 microcontroller, such as the Renesas RA4M1 found in the Arduino UNO R4 WiFi. For an edge intelligence unit to be viable for life-safety applications, it must maintain an inference latency of less than 100ms to ensure a near-instantaneous response to evolving fire threats (Das & others, 2025). This objective involves investigating model optimization techniques, including quantization and pruning, to balance classification accuracy with the memory and processing limits of the embedded platform.

Third, the research investigates the efficacy of a three-class classification paradigm (fire, no-fire, and false alarm) compared to traditional binary models. While most existing literature categorizes events simply as fire or no-fire, this binary approach frequently results in high

false-positive rates when the model encounters intense non-fire triggers like cooking fumes or cleaning sprays (Müller et al., 2024). By explicitly training the model to recognize the statistical signatures of these “false alarm” scenarios, this study seeks to enhance the precision of the actual fire classification and provide more actionable intelligence for building automation systems.

1.4.2 Scientific and Technical Contributions

This thesis provides several contributions to the field of intelligent fire safety and edge computing. A significant technical contribution is the development of a validated edge intelligence unit architecture that utilizes asymmetric multi-processing. In this configuration, the primary microcontroller manages real-time sensor acquisition and ML inference, while a dedicated co-processor handles the connectivity backbone for MQTT-based telemetry (Das & others, 2025). This modular design mimics established robotic controller architectures, ensuring that the critical “reflex” of fire detection remains autonomous and unaffected by network latency or connectivity failures.

Furthermore, this research contributes a unique multi-modal dataset specifically designed for three-class fire detection. Unlike many publicly available datasets that focus on visual fire detection or limited sensor arrays, the data generated in this study includes high-frequency (10 Hz) time-series measurements from five distinct sensing modalities across a wide range of real-world fire and false alarm scenarios. This dataset facilitates the analysis of class separability and the identification of “truth sensors”—such as carbon monoxide for combustion verification—that are essential for reducing false alarms in stationary autonomous nodes (Zheng et al., 2022).

Finally, the study provides a comprehensive performance benchmark of TinyML models on low-power Cortex-M4 hardware. By documenting the trade-offs between model complexity, RAM usage, and inference latency, this work offers a roadmap for the deployment of intelligent safety systems in environments where cost and power consumption are critical constraints. The results demonstrate that through careful optimization, it is possible to achieve high-accuracy, multi-class fire detection on hardware that traditionally lacked the computational capacity for advanced machine learning (Das & others, 2025).

1.5 Thesis Organization

The structure of this thesis is designed to provide a systematic progression from theoretical foundations to the empirical validation of an autonomous multi-sensor fire detection node. This logical flow is essential for addressing the complex interdependencies between hardware constraints, sensor modalities, and machine learning performance in safety-critical edge environments (Park et al., 2020).

The document is organized into ten chapters. Following this introductory chapter, **Chapter 2** presents a systematic literature review conducted using a PRISMA-inspired protocol to identify research gaps in false alarm reduction and TinyML deployment (Page & others, 2021). **Chapter 3** establishes the theoretical background, covering the physics of combustion, sensor fusion principles, and neural network optimization for microcontrollers. **Chapter 4** details the selection and characterization of the MEMS sensors used in the prototype. **Chapter 5** describes the hardware integration, specifically focusing on the asymmetric multi-processing architecture of the Arduino UNO R4 WiFi.

The experimental methodology is detailed in **Chapter 6**, outlining safety protocols and the rationale for the three-class classification strategy. **Chapter 7** describes the technical

implementation within the Edge Impulse platform and the development of the real-time inference firmware. **Chapter 8** evaluates the system through extensive experimental results, including an ablation study and on-device performance metrics. **Chapter 9** discusses the implications of class separability and the ethical considerations of AI in life-safety systems. Finally, **Chapter 10** summarizes the key findings, acknowledges limitations, and proposes avenues for future development, such as mesh network scalability.

2 Literature Review

2.1 Literature search and selection method (PRISMA-inspired)

This semester thesis uses a PRISMA-inspired rapid systematic literature review (SLR) protocol to ensure the literature review is transparent, repeatable, and aligned with the thesis research questions.

2.1.1 Scope and eligibility criteria

- **Publication window:** 2015–present
- **Domains:** indoor/building fire detection and wildland/wildfire detection
- **Methods covered:** multi-sensor approaches and sensor fusion families discussed in this thesis (e.g., Kalman filtering, Bayesian methods, neural networks, and hybrid strategies)
- **Language:** English
- **Evidence requirement:** the study must describe a method and provide at least one form of evaluable evidence (metrics, experiments, dataset/scenario description, or deployment constraints)

2.1.2 Information sources and search strategy

The search targets peer-reviewed journals and conferences indexed in IEEE Xplore, ACM Digital Library, and broad aggregators such as Scopus/Web of Science (when available). Google Scholar is used as a supplemental source for citation-based snowballing (forward and backward).

Two complementary queries are used to guarantee coverage of both domains:

- **Q1 (Indoor):** building/indoor fire detection AND multi-sensor/fusion AND (false alarms OR embedded/edge)
- **Q2 (Wildland):** wildfire/wildland/forest fire AND (fusion OR embedded/edge) AND (false alarms OR early detection)

The complete protocol, screening log template, extraction table template, and PRISMA flow template are stored in the thesis repository for auditability.

2.1.3 Screening, inclusion, and reporting

Records are deduplicated, screened by title/abstract, and then assessed at full text with exclusion reasons logged. The final included set is synthesized using a comparison table (sensors, fusion method, class framing, domain tag, and deployment feasibility) and a narrative review organized by: (i) indoor vs wildland evidence, (ii) fusion method families, (iii) false-alarm mitigation strategies, and (iv) edge feasibility.

3 Theoretical Background

4 Sensor Characterization

5 Hardware Integration

6 Data Collection

7 Implementation

8 Results Evaluation

9 Discussion

10 Conclusion

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